

Late Gate: Parametric Flight-Delay Stubs

Company technical paper · Version eng-lock.v1 **Status:** Internal Eng / Product **Pricing lock:** Takeoff $\pi = \$14$; Arrival $\pi = \$9$; $B(30/45/60) = 100/150 / \200 (both products). Fixed premium; scaled payout. τ internal only.

Key takeaways 1. Locked traveler prices are **\$14 Takeoff** and **\$9 Arrival**, with payouts 100/150 / \$200 at internal buffers 30 / 45 / 60. 2. US pool-average $P(\text{delay} \geq 30)$ is about **18%** (BTS Reporting Carrier OTP microdata, winter + summer month samples, $n \approx 1.14\text{M}$). At that rate, locked π loses money. 3. Positive house EV requires **selective underwriting**: only list stubs when flight-level $\hat{p}$ is under product $\times \tau$ gates (Arrival is stricter). 4. Clean demo prior **8% / 5% / 3.5%** restores Takeoff EV $\approx +6to+7$ and Arrival EV $\approx +1to+2$ per stub. 5. Scaling factors matter: summer-month sample shows HA $\approx 6.6\%$ vs AA $\approx 28.9\%$ $p \geq 30$ dep; morning $\approx 7.6\%$ vs evening $\approx 35.6\%$; SEA $\approx 14.5\%$ vs CLT $\approx 35.9\%$.

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1. At a glance

Late Gate sells limited-inventory **parametric stubs** on two outcomes per flight: **Takeoff delay** and **Arrival delay**. The traveler picks one delay-buffer tier. The stub pays a fixed cash amount if observed schedule delay meets or exceeds an internal threshold. The house is the sole counterparty in the MVP.

Variables (front-loaded)

Symbol	Meaning	Locked / default
π	Premium paid by traveler	Takeoff \$14 ; Arrival \$9
$B(\tau)$	Payout if delay $\geq \tau$	100/150 / \$200 at $\tau = 30 / 45 / 60$
τ	Internal threshold (minutes)	Never shown in traveler UI
p or $\hat{p}$	$P(D \geq \tau)$, true or estimated	Flight-level underwriting input
λ	Load = $\pi / E[\text{payout}]$	Target 1.45 on underwritten book
EV_house	$\pi - p \cdot B$	Must be > 0 on sold stubs

Decisions that make the model work

Decision	Why
Fixed π , scaled B	Simple UX; traveler picks Short / Medium / Long buffer
Selective underwriting	Locked π is underwater at pool-average p ($\sim 18\%$ at ≥ 30)
Target $\lambda = 1.45$	Includes $\sim 20\text{pp}$ adverse-selection / ops buffer vs fair odds
CUTOFF + HOT	Shrink purchase-timing adverse selection
Inventory caps	FOMO UX and bounded payout exposure
Gate-out / gate-in settlement	Matches BTS DepDelay / ArrDelay priors

Empirical snapshot

BTS Reporting Carrier OTP, winter month + summer month samples, operated non-cancelled non-diverted, $n \approx 1.14\text{M}$:

Product proxy	$P(\geq 30)$	$P(\geq 45)$	$P(\geq 60)$
Departure / Takeoff (DepDelay)	18.33%	13.61%	10.50%
Arrival (ArrDelay)	18.66%	13.83%	10.69%

At locked π and B , expected payout on a **random** US flight exceeds premium (house EV < 0). Positive house EV is restored on a **clean demo** / **selectively underwritten** book ($\hat{p}_{30/45/60} \approx 8\% / 5\% / 3.5\%$).

2. Locked product parameters

Variables callout Premium is fixed per product. Payout scales with the traveler-selected buffer. Threshold minutes stay internal.

Product code	Traveler name	Settlement metric	BTS proxy	Premium π
TAKEOFF	Takeoff delay	Gate-out – scheduled dep	DepDelay	\$14.00
ARRIVAL	Arrival delay	Gate-in – scheduled arr	ArrDelay	\$9.00

Internal τ (min)	UI label	Payout B
30	Short buffer	\$100
45	Medium buffer	\$150
60	Long buffer	\$200

Contract rule: Traveler selects **one** τ per stub. Pays B if observed delay_minutes $\geq \tau$, else 0 (single-tier, not cumulative).

Inventory	Default	Demo	Min	Max
Stubs per flight $\times$ product	10	5	3	20
Max payout exposure / flight $\times$ product	\$2,000	\$1,000	n/a	inventory $\times$ \$200

House is sole counterparty (no user LP) in MVP. Machine-readable twin: [rails-pricing-constants.json](#).

3. Industry baseline rates

3.1 US ≥ 15 minutes (published)

BTS reporting-carrier late arrivals/departures remain about **20-21%** at the ≥ 15 minute gate standard across recent annual summaries. Marketing-carrier arrival delay is in the same band. Treat $\approx 20\% \pm 1pp$ as the industry ≥ 15 backdrop. Exact $\geq 30/45/60$ come from microdata below (section 3.2), not from the ≥ 15 reporting standard.

3.2 US $\geq 30 / 45 / 60$ (this study)

Method: TranStats Reporting Carrier OTP for one winter month and one summer month; filter Cancelled=0, Diverted=0; share with DepDelay / ArrDelay $\geq \tau$.

Sample	Dep ≥ 30	Dep ≥ 45	Dep ≥ 60	Arr ≥ 30	Arr ≥ 45	Arr ≥ 60
Winter month	15.47%	11.27%	8.57%	15.94%	11.54%	8.78%
Summer month	20.77%	15.61%	12.15%	20.99%	15.80%	12.32%

Sample	Dep ≥30	Dep ≥45	Dep ≥60	Arr ≥30	Arr ≥45	Arr ≥60
Weighted	18.33%	13.61%	10.50%	18.66%	13.83%	10.69%

Artifacts: `data/empirical_delay_rates_*.csv`, `figures/empirical_delay_cdf.png`.

Caveats: Two months only; summer worse; cancellations/diversions excluded from the delay CDF (need separate rules); not conditioned on purchase timing (adverse selection).

3.3 Europe (published aggregates)

EUROCONTROL all-causes: arrivals within 15 min of STA around low-to-mid 70s% in recent annual digests (so >15 around mid-to-high 20s%). Monthly briefs publish delay bands (16-30 / 31-60 / >60). Example monthly shares delayed >30 min often sit in the mid-teens. Exact annual EU $p_{\geq 45}$ / $p_{\geq 60}$ as a single scalar was not extracted from materials reviewed. Treat as **unknown** pending CODA microdata or Cirium/OAG.

4. Scaling factors

Heterogeneity dominates pool averages. Summer-month departure sample ($p_{\geq 30}$) illustrates why selection works.

Scaling callout (summer month, dep $p_{\geq 30}$) Airline HA 6.6% vs AA 28.9%. Time block 0600-0659 7.6% vs 2000-2059 35.6%. Origin SEA 14.5% vs CLT 35.9%. DOW Thu 17.5% vs Fri 25.7%.

Split	Low example	High example
Airline	HA 6.6%	AA 28.9%
Dep time block	0600-0659 7.6%	2000-2059 35.6%
Origin (top volume)	SEA 14.5%	CLT 35.9%
Day of week	Thu 17.5%	Fri 25.7%

Full tables: `data/empirical_delay_splits_jul2024.csv` (filename retains sample label; treat as summer-month split file).

Product implication: Prefer morning banks, historically clean carriers/origins, and off-peak DOW for Arrival inventory (Arrival $p_{\max}$ gates are tighter than Takeoff). Origin airport is a practical **gate / airport proxy** for local congestion risk until gate-level telemetry is wired.

5. Mathematical model and decisions

5.1 Contract

For product $k \in \{\text{TAKEOFF}, \text{ARRIVAL}\}$ and traveler-selected internal threshold $\tau \in \{30, 45, 60\}$:

$$\text{Payout} = B(\tau) \cdot 1\{D_k \geq \tau\}$$

$$B(30)=100, B(45)=150, B(60)=200$$

Premium π_k is **fixed** per product (not per τ):

$$\pi_{\text{TAKEOFF}} = 14$$

$$\pi_{\text{ARRIVAL}} = 9$$

Expected payout: $E[\text{payout}] = p_k(\tau) \cdot B(\tau)$ where $p_k(\tau) = P(D_k \geq \tau)$.

5.2 Load and house EV

$$\pi = \lambda \cdot E[\text{payout}] = \lambda \cdot p \cdot B$$

$$\text{EV}_{\text{house}} = \pi - p \cdot B$$

Target $\lambda = 1.45$ on the underwritten book (includes ~20pp adverse-selection / ops buffer vs pure fair odds $\lambda=1$).

Equivalently, at locked π only sell when:

$$p \leq p_{\text{max}} = \pi / (\lambda \cdot B)$$

5.3 Why selective underwriting

Locked π and B are product/UX locks. They are **not** equal to $\lambda \cdot E[\text{payout}]$ at US pool-average p . At pool rates, implied λ is about **0.4-0.8** (house EV deeply negative). The noteworthy mathematical decision:

Keep locked π/B for traveler UX. Enforce λ via flight-level $\hat{p}$ gates (UNDERWRITE_REJECT). Do not sell random-pool flights at these prices.

5.4 Underwriting gates (p_{max} at $\lambda=1.45$)

Product	τ	$p_{\text{max}} (\lambda=1.45)$	Break-even $p (\lambda=1)$
Takeoff	30	9.66%	14.00%
Takeoff	45	6.44%	9.33%
Takeoff	60	4.83%	7.00%
Arrival	30	6.21%	9.00%
Arrival	45	4.14%	6.00%
Arrival	60	3.10%	4.50%

5.5 Delay definitions → products

Industry metric	Definition	Late Gate product
Gate departure delay	Actual gate-out – scheduled dep	Takeoff delay
Wheels-off delay	Runway liftoff – schedule	Not primary trigger
Gate arrival delay	Actual gate-in – scheduled arr	Arrival delay
BTS "late flight"	≥15 minutes gate late	Reporting standard only

Recommendation: Settle Takeoff on **gate-out vs published STD**; Arrival on **gate-in vs published STA**. Traveler-facing "Takeoff delay" means departure lateness from schedule, not wheels-up. **Never display τ minutes in UI.**

5.6 Alternatives if Eng reopens pricing (not locked)

For an **unselective** US pool book at $\lambda=1.45$ and current B: $\pi \approx 27-31$ both products. Or keep π and reduce B to about **30–55** (poor UX). **Recommendation:** keep locked π/B + selection for demo; revisit risk-based π later.

6. House EV under pool vs clean prior

6.1 Pool-average book (underwater)

Weighted winter + summer microdata:

Product	τ	p	E[payout]	π	House EV	Implied λ
Takeoff	30	18.33%	\$18.33	\$14	–\$4.33	0.76
Takeoff	45	13.61%	\$20.41	\$14	–\$6.41	0.69
Takeoff	60	10.50%	\$21.00	\$14	–\$7.00	0.67
Arrival	30	18.66%	\$18.66	\$9	–\$9.66	0.48
Arrival	45	13.83%	\$20.75	\$9	–\$11.75	0.43
Arrival	60	10.69%	\$21.38	\$9	–\$12.38	0.42

6.2 Clean demo prior (recommended)

Assume selectively listed flight with $\hat{p} = \{8\%, 5\%, 3.5\%\}$ at $\{30, 45, 60\}$:

Product	τ	E[payout]	House EV	λ
Takeoff	30	\$8.00	+\$6.00	1.75
Takeoff	45	\$7.50	+\$6.50	1.87
Takeoff	60	\$7.00	+\$7.00	2.00
Arrival	30	\$8.00	+\$1.00	1.12

Product	τ	E[payout]	House EV	λ
Arrival	45	\$7.50	+\$1.50	1.20
Arrival	60	\$7.00	+\$2.00	1.29

Arrival at \$9 is tight ($\lambda \approx 1.1$ -1.3 on this prior). Prefer morning / historically clean flights for Arrival inventory.

6.3 Scenario ladder (research)

Scenario	$\hat{p}_{30/45/60}$	Takeoff EV @ τ_{30}	Arrival EV @ τ_{30}
ultra_clean	5 / 3 / 2%	+\$9.00	+\$4.00
clean_demo	8 / 5 / 3.5%	+\$6.00	+\$1.00
HA-like (summer dep)	6.6 / 3.6 / 2.4%	+\$7.40	+\$2.40
AS-like	13.3 / 7.9 / 5.2%	+\$0.70	-\$4.30
pool_avg	~18.5 / 13.7 / 10.6%	loss	loss

See `data/pricing_selective_scenarios.csv` and `figures/house_ev_sensitivity.png`.

7. Underwriting gates and risk controls

Risk callout Without CUTOFF, HOT, and $p_{\max}$ gates, locked π is insufficient even on "clean" scheduled flights because purchase timing correlates with delay information.

7.1 Refusal codes

Code	Rule
HOT	Freeze new sales if delay already developing, weather watch, or soft $\hat{p}$ warn
CUTOFF	No sales inside 4h of STD (demo 6h)
FULL	Inventory exhausted
DUPLICATE	Max 1 open stub per user per flight $\times$ product
UNDERWRITE_REJECT	$\hat{p} > p_{\max}$ for product $\times\tau$
EXPOSURE_CAP	Portfolio / cluster limit hit
CANCELLED	MVP: refund premium
DIVERTED	Arrival: pay max B or review

7.2 Portfolio / weather caps

Limit	USD
Max net payout exposure / airport-day	25,000
Max net payout exposure / airline-day	40,000
Max net payout exposure / book-day	150,000
Correlation cluster cap	15,000
Weather-watch airport remaining-capacity multiplier	0.35

Prefer diversifying away from single-hub evening banks.

7.3 Traveler-safe wording

Avoid (UI)	Prefer
insurance, policy, claim, premium	stub, locked price, payout
gamble, bet, odds	delay cushion, buffer tier
show "30/45/60 minutes" as τ	Short / Medium / Long buffer
house edge, λ , $\hat{p}$	Limited stubs on this flight

Paper/actuarial language is fine internally. Traveler surfaces should stay assistance/ancillary-protection tone, without claiming regulated insurance unless licensed.

7.4 Future depositor pool (optional)

MVP: house sole counterparty. Future sketch: depositors post capital; after traveler payouts and ops, residual edge $(\lambda-1) \cdot E[\text{payout}]$ can be shared (example split 40% house ops / 50% depositor yield / 10% reserve). Depositors inherit house edge only after underwriting and reserves. They do not remove the need for $p_{\max}$ gates. No user LP in current product.

8. Settlement definitions

1. **Oracle:** Production: Cirium or OAG (or airline direct). MVP: partner status API; BTS for offline backtest only (lag).
2. **Metric:** Takeoff = gate-out delay; Arrival = gate-in delay vs published times at purchase lock.
3. **Observe:** After actual event or scheduled time + τ + **30 min** buffer.
4. **Auto-pay** if high-confidence source and $D \geq \tau$; else hold.
5. **Dispute window:** 72 hours.
6. **Cancelled:** refund π (MVP). **Diverted:** Arrival special case.

9. Data sources

Primary free US prior: BTS Airline On-Time Performance (Reporting Carrier). Europe: EUROCONTROL CODA digests and monthly briefs. Commercial settlement candidates: Cirium Sky API, OAG Flight Status. US airport/carrier metrics: FAA ASPM / ASQP. ADS-B tracking (OpenSky) is **not** CRS schedule delay.

Labeled links live in [References](#). Local study artifacts: [data/](#), [figures/](#).

Blockers: Cirium/OAG paywalled (needed for live settlement). BTS Excel summary tables often blocked to datacenter IPs (prezip CSVs work). OpenSky unsuitable as sole settlement oracle for schedule delay. Full-year microdata not fully ingested here (two-month sample).

10. Sensitivity and assumptions

10.1 House EV vs p at locked π

See [figures/house_ev_sensitivity.png](#). Zero-crossing at break-even $p = \pi/B$. Pool p_{30} ($\approx 18\text{-}19\%$) lies deep in the loss region for both products.

10.2 Inventory $\times$ premium income

Inventory	Takeoff premium income	Max exposure ($\tau=60$)	Arrival premium income
5	\$70	\$1,000	\$45
10	\$140	\$2,000	\$90
20	\$280	\$4,000	\$180

House EV scales with $\Sigma_i (\pi - p_i B_i)$ over sold stubs. Negative if p_i uncontrolled.

10.3 Traveler fair-odds comparison (clean demo prior)

For Takeoff $\tau=30$: fair premium ≈ 8.00 ; *travelerpays14* $\rightarrow$ load 1.75. For Arrival $\tau=30$: fair ≈ 8.00 ; *pays9* $\rightarrow$ load 1.12 (thin). Communicate value as **certainty of payout rules + limited inventory**, not "cheap odds."

10.4 Assumptions (labeled)

- **A1:** Settlement equals BTS-style gate delay (assumption until Cirium/OAG wired).
- **A2:** Sold-stub p equals historical flight-level $\hat{p}$ if CUTOFF/HOT enforced (optimistic without live telemetry).
- **A3:** Single-tier payout (not stacked 30+45+60).
- **A4:** Cancellations refund π (not a delay win).
- **A5:** Clean demo prior 8/5/3.5% is a **scenario**, not a national statistic.
- **A6:** Full-year microdata not fully ingested. Expand before production rate cards.

10.5 Rails constants

See [rails-pricing-constants.md](#) and [rails-pricing-constants.json](#) for drop-in thresholds, premiums, payouts, inventory, λ , refusal codes, and exposure limits.

11. References

1. [BTS Airline On-Time Performance Database](#)
2. [BTS Download \(Reporting Carrier\)](#)
3. [BTS TranStats annual charts](#)
4. [BTS Marketing Carrier Annual](#)
5. [BTS on-time gallery](#)
6. [EUROCONTROL CODA Digest \(annual all-causes delays\)](#)
7. [EUROCONTROL monthly delay brief \(example PDF\)](#)
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13. [OpenSky terms of use](#)
14. [Wang et al. delay CCDF \(arXiv HTML\)](#)
15. [AXA HK Express parametric protection](#)
16. [Cover Genius Delay Valet](#)
17. [Blink Flight Disruption](#)
18. [Hopper / OAG case study](#)
19. Late Gate empirical outputs: local `data/` and `figures/` under this docs pack

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